Constellations: From Ancient Stories to the Modern 88

What you will be able to do

1. Define a **constellation** in both the old (pattern) and the modern (IAU region) sense.
2. Trace the history: Mesopotamia → Greece (Hipparchus, **Ptolemy**) → the Islamic world → India → the age of exploration → the **IAU's 88**.
3. Read **Bayer** (Greek-letter) and **Flamsteed** (number) star names.
4. Recall famous constellation stories from Greek and Indian traditions.

1. What is a constellation?

Old meaning: a **pattern of stars** imagined as a figure — a hunter, a scorpion, a plough, a deer. **Modern meaning (since 1930):** a precisely bounded **region of the sky**. The International Astronomical Union (IAU) divides the whole celestial sphere into **88 constellations**, like countries on a world map; every point of the sky belongs to exactly one.

Important

The stars of a constellation are usually **not** physically related — they just lie in the same direction. In Orion, Bellatrix is ~250 light-years away and Alnilam ~2,000. Seen from another star system, the patterns would look completely different.

2. One sky, many pictures

Every culture drew its own pictures on the same stars. Here is the region around Orion and Taurus as drawn by six traditions, rendered by Stellarium on the real sky:

3. A short history of constellations

Mesopotamia (Babylon and Sumer), before 1000 BCE

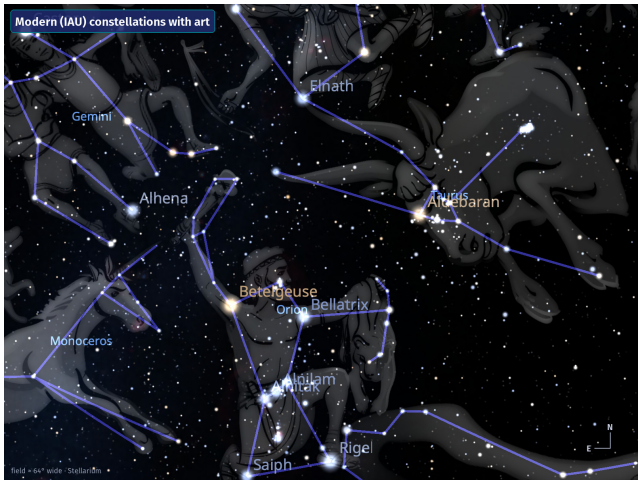
Clay tablets called **MUL.APIN** (~1000 BCE) list about 66 stars and constellations, including the Bull (Taurus), the Lion (Leo), the Scorpion and the Goat-Fish (Capricornus), several of which we still use. The Babylonians also divided the ecliptic into **12 equal signs** — the origin of the zodiac.

Ancient Greece

- **Eudoxus** (~4th century BCE) described the constellations; the poet **Aratus** (~270 BCE) turned them into the poem *Phaenomena*.
- **Hipparchus** (~190–120 BCE) made the first big star catalogue (~850 stars), graded stars into six brightness classes, and **discovered precession** (Lecture 1.2).

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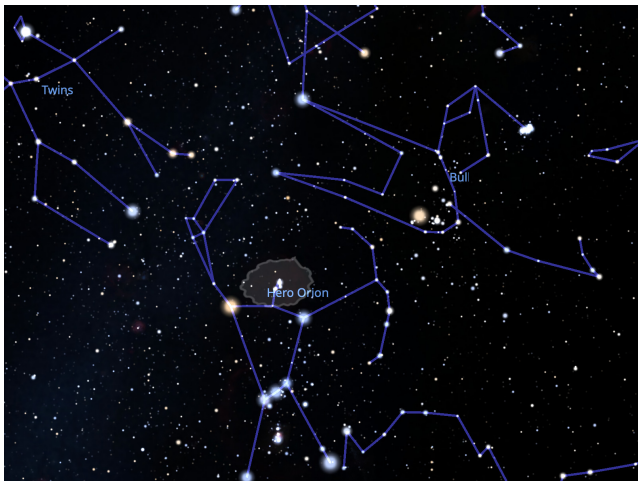
P.T.O.



Modern IAU constellations
art by Johan Meuris, Free Art License



Babylonian MUL.APIN (~1000 BCE)
paintings by Jessica Gullberg; S. M. Hoffmann; CC BY-ND 4.0



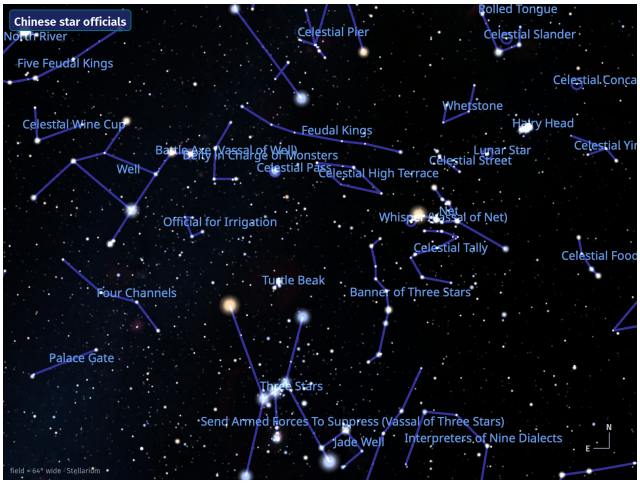
Ptolemy's *Almagest* (~150 CE)
Stellarium "Greek Almagest" sky culture, CC BY-ND 4.0



Al-Sufi's *Book of Fixed Stars* (964 CE)
compiled by Khalid Al-Ajaji, CC BY-ND 4.0



Indian nakshatras
S. M. Hoffmann with B. S. Shylaja, CC BY 4.0



Chinese star officials (*xingguan*)
Stellarium "Chinese" sky culture, CC BY-SA 4.0

The same patch of sky around Orion and Taurus as drawn by six traditions — screenshots from Stellarium. Orion's Belt, the Pleiades and the V of Taurus appear in almost every tradition.

Ptolemy's *Almagest* (about 150 CE): the 48 classical constellations

Claudius Ptolemy of Alexandria listed 1,022 **stars in 48 constellations**. For 1,400 years the *Almagest* was the standard astronomy textbook in Europe and the Islamic world, and **all 48 constellations survive** among today's 88 (one, the ship Argo Navis, was later split into three).

The Islamic golden age (8th–15th centuries)

Ptolemy's work was translated into Arabic and improved. **Al-Sufi's *Book of Fixed Stars*** (964 CE) combined Greek figures with Arabic star lore and illustrated each constellation twice — as seen on a globe and as seen in the sky. That is why so many bright stars have **Arabic names** (Lecture 1.5).



al-Sufi, *Book of Fixed Stars* (copy in the BnF)



J. Bayer, *Uranometria* (1603)



S. Hall, *Urania's Mirror* (1824)



J. Hevelius, Ursa Major (1690)

Constellation art through the ages: three historic pictures of Orion, and the Great Bear of Hevelius.

All public domain, via Wikimedia Commons.

India: nakshatras, Saptarishi and rashis

- The **nakshatras** (Lecture 1.4) — 27 or 28 star-groups along the Moon's path — appear in the Vedic literature, and the *Vedanga Jyotisha* used them for time-keeping.
- **Saptarishi Mandala** — the seven sages — is the Big Dipper; **Dhruva** is the Pole Star.
- The **12 rashis** reached India through contact with the Hellenistic world in the early centuries CE (e.g. the *Yavanajataka*, 2nd century CE). Their Sanskrit names translate the Greek ones: Mesha = ram, Vrishabha = bull, Mithuna = couple, Simha = lion...

China

Chinese astronomers developed an independent system of **283 asterisms** with **28 lunar mansions** (*xiu*) — very similar in idea to the nakshatras.

The age of exploration: filling the southern sky (1590s–1750s)

Ptolemy's Alexandria (31° N) could not see the far southern sky. Navigators sailing to the East Indies filled the gap:

Who	When	Added (examples still used today)
Keyser & de Houtman (Dutch navigators), published by Plancius and in **Bayer's *Uranometria*** (1603)	1595–1603	12 southern constellations: Pavo (Peacock), Tucana, Grus, Phoenix, Dorado, Volans, Musca, Apus, Chamaeleon, Hydrus, **Indus**, Triangulum Australe
Johannes Hevelius (Poland)	1690	7 still used: Lynx, Canes Venatici, Lacerta, Leo Minor, Scutum, Sextans, Vulpecula

(continued)

Who	When	Added (examples still used today)
Nicolas-Louis de Lacaille (France, from the Cape of Good Hope)	1750s	14 named after instruments: Telescopium, Microscopium, Antlia, Pyxis, Fornax, Horologium, Octans, Sculptor, Norma, Circinus, Reticulum, Caelum, Pictor, Mensa; he also split **Argo Navis** into Carina, Puppis and Vela

Fun fact

The constellation **Indus** ("the Indian") was added by the Dutch navigators in the 1590s, and **Pavo**, the Peacock, may have been inspired by peacocks seen in South and South-East Asia.

The IAU fixes the sky (1922–1930)

By 1900 there were rival maps and made-up constellations. The newly formed IAU

- **1922 (Rome):** adopted **88 constellations**, each with a standard 3-letter abbreviation (Ori, UMa, Sco, Cru…);
- **1930:** published exact **boundaries**, drawn by the Belgian astronomer **Eugène Delporte** along lines of RA and Dec.

Largest: **Hydra** (1,303 square degrees). Smallest: **Crux** (68 square degrees). 48 come from Ptolemy; 40 were added in the 16th–18th centuries.

4. How stars are named

Proper names. About 450 stars have traditional names, mostly Arabic, Greek or Latin, standardised since 2016 by the IAU Working Group on Star Names.

Bayer designations (1603). Greek letters, *roughly* in order of brightness, plus the Latin genitive of the constellation:

Letter	Usually	Example
α alpha	the brightest	α Orionis = Betelgeuse
β beta	second	β Orionis = Rigel (actually brighter than α!)
γ gamma	third	γ Orionis = Bellatrix
$\delta, \varepsilon, \zeta, \eta, \theta \ldots$		δ Orionis = Mintaka

Genitives: Orion → *Orionis*, Scorpius → *Scorpii*, Canis Major → *Canis Majoris*, Lyra → *Lyrae*, Ursa Major → *Ursae Majoris*, Centaurus → *Centauri*, Crux → *Crucis*, Cygnus → *Cygni*.

Flamsteed numbers (1725). Numbered from west to east (in order of RA), e.g. **61 Cygni** — the first star whose distance was measured (1838) — and **51 Pegasi**, the first Sun-like star found to have a planet (1995). Fainter stars carry only catalogue numbers.

5. Stories in the stars

Orion and the Scorpion (Greek). The boastful hunter Orion was stung by a giant scorpion; both were placed in the sky on **opposite sides**. Orion (RA $\sim 5.5^{h}$) sets as Scorpius (RA $\sim 16.5^{h}$) rises: the hunter forever flees the scorpion.

Indian sky stories

Mriga and Lubdhaka. In the *Aitareya Brahmana*, Prajapati took the form of a deer (**Mriga**, the Orion region) and was shot by an arrow — the three **belt stars** (*Ishu*) — fired by the hunter **Lubdhaka** (**Sirius**, also *Mrigavyadha*). The belt really does point at Sirius!
Saptarishi and Arundhati. The seven stars of the Big Dipper are the sages **Kratu, Pulaha, Pulastya, Atri, Angiras, Vashishtha, Marichi**. Beside Vashishtha (Mizar) is a faint companion, **Arundhati** (Alcor). Couples are traditionally shown Arundhati at weddings — and spotting Alcor is an ancient **eyesight test**.
Trishanku. King Trishanku, sent upward by the sage Vishwamitra and pushed back by the gods, hangs in a heaven of his own — identified with **Crux**.
Agastya. Canopus, the second-brightest star, bears the name of the sage who travelled south. Its first dawn appearance (*Agastya udaya*) was linked with the end of the monsoon.

Cassiopeia's family (Greek). Queen **Cassiopeia** boasted of her beauty; her daughter **Andromeda** was chained to a rock for the sea monster **Cetus** and rescued by **Perseus** on the winged horse **Pegasus** — six constellations, one story.

Key points to remember

- Modern constellations are **88 IAU regions** (adopted 1922; boundaries 1930, Delporte).
- **Ptolemy's *Almagest*** (~150 CE): 48 constellations, 1,022 stars — all still used.
- Southern constellations: Dutch navigators (Bayer 1603), Hevelius (1690), Lacaille (1750s).
- Bayer letters ($\alpha, \beta, \gamma \ldots$) roughly follow brightness; Flamsteed numbers follow RA.
- Indian sky lore: nakshatras, Saptarishi & Arundhati, Dhruva, Trishanku, Agastya, Mriga & Lubdhaka.
- A constellation's stars lie at very different distances — it is a line-of-sight pattern.

Check yourself

1. How many constellations did Ptolemy list? How many are recognised today?
2. What does "β Orionis" mean? Which star is it?
3. Which ancient constellation was split into Carina, Puppis and Vela, and by whom?
4. Name the seven sages of the Saptarishi in order and say where Arundhati is.

Answers: 1. 48; 88. 2. The "beta" star of Orion — Rigel. 3. Argo Navis, by Lacaille. 4. Kratu, Pulaha, Pulastya, Atri, Angiras, Vashishtha, Marichi; Arundhati (Alcor) is beside Vashishtha (Mizar).

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §2.2 "Ancient Astronomy" — `https://openstax.org/books/astronomy-2e/pages/2-2-ancient-astronomy`
- IAU, "The Constellations" (official charts and names) — `https://www.iau.org/public/themes/constellations/`
- **Stellarium**: change the *sky culture* (sky & viewing options) to compare Indian, Arabic, Chinese and other traditions on the real sky.
- H. A. Rey, *The Stars: A New Way to See Them* (Houghton Mifflin).